Sisir Kausthubh

Applied ML | Human-in-the-loop AI | Retrieval & RAG | LLM Agents | LLM Evaluation | Vision-Language Models | Indic NLP & Speech

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SUMMARY

I build LLM and vision-language systems that work with people: retrieval that cites its sources and says when the answer is not there, agents that know when to hand a case to a person, and evaluations scored on held-out data next to the published baseline. I ship end-to-end products (models, APIs, UIs, integrations) quickly on one laptop GPU, and report accuracy, cost and failure modes together.

PROJECTS: RETRIEVAL, AGENTS & HUMAN-IN-THE-LOOP

docs-rag-assistant | chat with company filings, with page citations Fully local

- **61.8% on FinanceBench** with all 84 filings in one index (paper's GPT-4-Turbo RAG: 19%); 66.7% with the question's filing selected (50%).
- **98% correct refusals** when the filings are removed; every citation checked against its source page in code; one search per needed figure added +6.9 pts on test.
- Docling parsing, Qdrant retrieval, Qwen3 reranker, Gemma 4 26B on vLLM, FastAPI + Gradio UI.

catalog-automation-agent | agent with human review (LangGraph, MCP server, n8n)

- Fetch, tag, then **auto-approve or pause for a reviewer**: 79.2% of 1,610 products auto-tagged at **94.6% accuracy**; the rest routed to a person.
- QC mode catches 67.8% of wrong listings at 90% precision. Durable agent state: **0 lost, 0 duplicate writes** under injected failures and a killed process.
- Plugs into the tools people already use: Shopify, Slack, Google Sheets, n8n, REST/CSV, and Claude over MCP; 45 products/s with FP8.

document-extraction-pipeline | receipts and GST invoices to checked data

- 84.0 field F1 on CORD (Donut: 83.3, same scorer); 84% / 91% of invoices fully correct on seen / never-seen layouts.
- GSTIN, tax and PO checks decide what skips a person: 84-91% auto-accepted with **0 wrong ones let through**; 76 / 76 planted problems caught.

PROJECTS: EVALUATION & MEASUREMENT

- **visual-product-tagging:** 4B VLM LoRA-tuned in 48 minutes on one GPU; **85.3% on ImplicitAVE** with hand-checked labels (GPT-4V: 86.8%), +17.5 pts over the paper's best fine-tuned model, INR 0.67 per 1,000 products; confidence threshold auto-tags 79% at 94.6% and flags the rest.
- **efficient-smolvla-voice-to-action:** 7.7x faster robot-policy inference (111 ms to 14.4 ms median) with task success held at 82.00% over the same 400 closed-loop episodes, tested with a **paired McNemar test**; rejected FP8 vision after success fell to 80%.
- **Open source & talks:** reviewed and benchmarked Hugging Face LeRobot PR #4634 (2.2x end to end; the author added my numbers to the PR); lightning talk at the NVIDIA Jetson AI Lab Research Group call, 10 Nov 2026.

PROJECTS: LANGUAGE, SPEECH & MULTILINGUAL AI | BTech research

- **Tamil-to-Telugu speech-to-speech translation:** Whisper ASR (best of 4 models on Common Voice 11), IndicTrans2 translation, VITS/MMS speech synthesis.
- **Tamil summarization:** three extractive methods vs fine-tuned mBART/mT5, compared with ROUGE. **Malayalam headline generation** with a from-scratch Transformer.
- **Tamil multimodal hate-speech detection** (final-year research): IndicBERT text + ViT/ResNet video features, pseudo-labelling and synthetic data. Indic document OCR tooling.
- **LLM fine-tuning:** Llama 3.2 3B with QLoRA (Unsloth, TRL), trained on assistant replies only, exported to GGUF for local use.

OTHER PROJECTS

- **AI short-video generator** (Next.js, Remotion): topic to script, images, narration and word-timed captions in one product workflow.
- **Retail sales forecasting:** chronological folds and a separate holdout; Random Forest cut error 11% vs a seasonal baseline. **Warehouse navigation** with deep RL (DDPG, TD3) in Gazebo.

SKILLS

Retrieval & data: RAG, Qdrant, rerankers, Docling, citation checking, Spark

Agents: LangGraph, MCP, n8n, human-review routing, confidence thresholds, checkpointed/durable agent state

Evaluation: held-out splits, published baselines, hand-checked labels, refusal/abstention tests, paired statistics (McNemar), pytest, GitHub Actions

Models & inference: Python, PyTorch, Hugging Face (Transformers, TRL, LeRobot), LoRA/QLoRA, vLLM, FP8/INT8, torch.compile, TensorRT

Product: FastAPI, Gradio, Next.js, Docker, Linux, Git, uv

EDUCATION

BTech, Computer Science and Engineering (AI) | Amrita Vishwa Vidyapeetham Sep 2021 - Jun 2025